

EDUCATION

Madras Institute of Technology, Anna University	Chennai, Tamil Nadu, India
Master of Engineering in Aeronautical Engineering – CGPA 7.7	Sep 2025-Jun 2027
Rajalakshmi Engineering College	Chennai, Tamil Nadu, India
Bachelor of Engineering in Mechanical Engineering - CGPA: 8.5	Oct 2021-Apr 2025
International Indian School	Al-Jubail, Kingdom of Saudi Arabia
HSC: 88.20 %	2021
SSLC: 78.40 %	2019

SKILLS SUMMARY

- **CAD Software:** AutoCAD 2D, SolidWorks, NX Siemens, Fusion 360
- **Simulation Software:** Ansys Workbench, COMSOL Multiphysics, Ansys STK
- **PLM Software:** Windchill PLM, Teamcenter
- **Languages:** Python with Machine Learning, C Language, SQL
- **Data Visualization & Analysis Tools:** Power BI, Microsoft (Excel, PowerPoint, Word)
- **Soft Skills:** Rapport Building, Stakeholder Management, Excellent Communication
- **Personal Attributes:** Emotional Intelligence, Resilience, Attention to Detail, Problem Solving

WORK EXPERIENCE

Physical Functional Design & Integration – Engineering Trainee	August 2024 – January 2025
STELLANTIS FCA Engineering India Private Limited	Thoraipakkam, Chennai, Tamil Nadu, India
• Assisted in designing and validating physical components within vehicle systems, gaining hands-on experience with CAD modeling and simulation studies. Collaborated with cross-functional teams to understand the integration of mechanical, electrical, and software systems. Developed knowledge of automotive design standards, manufacturing feasibility, and quality requirements.	
Lignite Handling System – Internship Trainee	June 2023
NLC India Limited, Thermal Power Station II Expansion	Neyveli, Tamil Nadu, India
• Completed training on lignite extraction, crushing, transportation, and storage, gaining hands-on exposure to material-handling systems such as conveyors, crushers, stackers, and reclaimers; observed PLC/SCADA-driven automation; learned safety protocols, dust control measures, and environmental compliance requirements; and acquired foundational knowledge of large-scale material handling and power generation logistics.	
Refrigeration & Air Conditioning – Intern	January 2023
Spark Technical Services	Tambaram sanatorium, Chennai, Tamil Nadu, India
• Gained theoretical proficiency in vapour-compression cycles and HVAC design via component analysis; applied core thermodynamics and gas-law principles to model and optimize refrigeration cycles; conducted systematic virtual troubleshooting of AC units—diagnosing power, compressor, expansion valve, sensor, and refrigerant issues; and developed expertise in HVAC validation and oil management practices, including system zoning and lubricant selection.	

TRAININGS

➤ Data Analytics for Manufacturing	March 2024
TVS Training and Services Ltd	Chennai, Tamil Nadu, India
➤ Product Lifecycle Management Windchill	October 2024
Sumedhas Tech Solutions Pvt Ltd	Chennai, Tamil Nadu, India
➤ FIP-Fuel Injection Pump	October 2023
BOSCH Training Lab	Chennai, Tamil Nadu, India
➤ KTS-Vehicle Diagnosis System	October 2023

RESEARCH & ENGINEERING PROJECTS

Project N.E.S.A: Embedded Spatio-Temporal GNN & EHD Plasma Flight Control Framework 2026

- Served as R&D Lead developing a cyber-physical UAV flight control system that combines a PyTorch Geometric dual-headed Spatio-Temporal Graph Neural Network (50 nodes X 6 features X 10 timesteps, 0.006245 MSE loss) with high-voltage solid-state Dielectric Barrier Discharge (DBD) plasma actuators. Validated via OpenFOAM and ANSYS CFD sweeps that active electro-fluidic momentum injection yields a 290% retention gain in aerodynamic lift-to-drag (L/D) efficiency across transient stall boundaries at 25–40 m/s.

Design & Computational Analysis of Thermodynamic Baffles for Aviation Liquid Hydrogen Tanks 2026

- Engineered an internal G-10 CR fiberglass epoxy composite Thermal Dam to decouple mechanical slosh suppression from conductive heat transfer, eliminating the traditional metallic heat fin paradox. Developed a 2D Volume of Fluid (VOF) multiphase model with Lee Phase Change mechanics and k-omega SST damping in ANSYS Fluent to simulate take-off kinematics. Optimized a 5-station, 15% blockage perforation layout that reduced transient impact forces by 75% and verified cryogenic structural integrity via a 1-Way ANSYS Mechanical FSI analysis with a Factor of Safety of 32.55.

Hybrid Digital Twin for Aero-Engine Bearing Prognostics (AeroCON 2026 Paper) 2026

- Designed a deep-fusion Hybrid Digital Twin utilizing a Stacked LSTM trained on a dual-path unified loss function embedding the Paris-Erdogan Law to eliminate unphysical prediction errors and reduce RMSE by 71%. Generated 200,000+ synthetic profiles via Monte Carlo simulation to bridge the Sim-to-Real gap, validated the framework via Transfer Learning on the NASA IMS dataset with a 15% calibration baseline, and optimized the network for an edge-ready latency of 63.26 ms with Monte Carlo Dropout 95% confidence intervals.

Multi-Agent Aerodynamic Digital Twin & Physics-Informed Deep Learning Framework 2026

- Built an automated aerodynamic optimization pipeline in Python using a 5-chromosome Genetic Algorithm and XFOIL viscous solver loop to parameterize an NREL S809 airfoil via Hicks-Henne shape functions. Trained a deep PyTorch MLP data engine to achieve real-time geometric inversion ($C_p \rightarrow Y$, validation MSE: 9.37×10^{-6}), integrated an 11-point Savitzky-Golay filtering routine for numerical stability, and deployed a 4-agent distributed prognostic system to track physical surface roughness up to 88.9%

Aero-Scribe: Multi-Modal AI Agent for Optimized Aircraft Maintenance 2026

- Architected and developed a multi-modal, multi-agent AI system utilizing advanced Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) to seamlessly parse and extract actionable maintenance insights from high-volume technical documentation and MRO logs. Implemented predictive data analytics pipelines to evaluate sensor telemetry and historical records for early component degradation detection, managing the end-to-end version-controlled infrastructure deployment via GitHub.

Development of a Compact, Self-Deploying Satellite for Space Debris Collection 2025

- Developed a compact, self-deploying satellite system for space debris collection using a motorized Aerogel net and autonomous guidance. Integrated microcontrollers, sensors, and mechanical blades to capture and secure debris of various sizes in LEO. Designed for reusability and safe Earth re-entry for debris recovery and analysis. Focused on sustainable, low-cost orbital cleanup.

Design and Development of All-Terrain Vehicle BAJA SAEINDIA 2024

- Collaborated in the design, simulation, fabrication, and testing of a hydrogen-compatible All-Terrain Vehicle (ATV) for the BAJA SAEINDIA HBAJA 2024 competition. Contributed to optimizing mechanical, electrical, and thermal systems for performance and safety. Successfully competed in dynamic events, applying real-world engineering and project management skills.

Multi-metal 3D Printing for Complex Components 2024

- Developed a multi-metal 3D printing system capable of sequential and parallel deposition of Aluminum 6061, Copper C11000, and AISI 316L Stainless Steel using a coaxial powder feed and laser fusion. Integrated thermal cameras, wettability modifiers, and nickel transition layers to ensure strong intermetallic bonding and

reduce thermal shock. Post-processed parts with HIP and annealing to enhance mechanical strength and structural integrity.

Swachh Smart: Smart Highway Amenities Database (NHAI Project)

2024

- Conceptualized and built a centralized facility management platform for the National Highways Authority of India (NHAI) using a Python backend and MariaDB database layer. Formulated robust data ingestion pipelines to process live IoT sensor inputs, driving an AI analytics engine to deliver regional and national dashboards, predictive cleaning schedules, automated fault detection, and secure UPI payment streams.

Safe Journeys AI: The Walker App

2023

- Designed and developed an AI-powered personal safety mobile network and digital buddy system for student transit, engineering predictive analytics models for dynamic route safety scoring and real-time location tracking. Programmed custom anomaly detection algorithms to identify path deviations for automated emergency triggering, integrating a crowdsourced incident-reporting interface with gamified community hazard mapping.

Development of Portable Water Sterilization Unit

2023

- Designed and fabricated a zero-electricity, portable water sterilization unit using friction-generated heat for rural and emergency use. Integrated multi-stage filtration with a copper cylinder-based frictional heating mechanism. Employed SolidWorks modeling, CNC fabrication, and 3D printing for cost-effective and durable construction.

Determining Porosity of Rubber Products

2023

- Developed a non-destructive testing method using ultrasonic sensors to detect porosity in rubber tyres, replacing traditional invasive inspection techniques. The system generated C-scan visualizations from time-of-flight data, enabling accurate 2D/3D mapping of internal pores to enhance tyre safety and structural assessment.

Advanced Pothole Detection and Mitigation System

2023

- Designed a real-time pothole detection system using LIDAR, ultrasonic sensors, and 3D mapping, integrated with a cloud-based platform and mobile app to deliver proactive driver alerts 30–40 meters in advance. The system leveraged crowdsourced data and Google Maps API to enhance road safety, hazard reporting, and route optimization.

Advanced Storage Unit for Tomatoes

2023

- Designed an IoT-enabled, thermally controlled tomato storage system integrating UV-C sterilization, humidity and ethylene management, and food-grade wax coating to extend shelf life. Developed a scalable, zero-waste model incorporating activated charcoal and sensor-based monitoring for optimal post-harvest preservation. Focused on cost-effective implementation using renewable energy and smart automation.

CERTIFICATES

- ❖ NPTEL Certificate on Joining Technologies for Metals

❖ NPTEL Certificate on Social Innovation and Industry 4.0

❖ NPTEL Certificate on Product Design and Manufacturing 2023
- October 2024

October 2023

April 2023

ACHIEVEMENTS

- ❖ Authored and published a peer-reviewed **research paper** on Hybrid Digital Twins and physics-informed machine learning for aero-engine prognostics in the *SAE International Research Journal*.

❖ Secured a formal **design patent** for a novel, compact, self-deploying satellite architecture optimized for orbital space debris collection in Low Earth Orbit (LEO).

❖ Crowned Microsoft AI-Odyssey champion for deploying optimized cloud AI models in Bangalore

❖ Recognized as a top regional finalist in the ICT Academy Youth Talk platform in the Chennai Region for public speaking and technical communication.
- 2026

2025

2024

2024